PulseIQ | Phase 1 Report

Aura Workforce by Growmar - Data Analytics Track - Foundation

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Repository	Private GitHub: main published	Phase tag	phase-1 at 965d9f6

Phase 1 built a validated local warehouse, a 28-metric governed layer, and the first workforce dashboard. The private GitHub repository and phase tag are published. Public hosting and programme submission still need the account and owner actions listed on page 2.

Five task outcomes

Task	Result	Evidence
1. ETL and contracts	Complete locally	29 CSV feeds; row counts reconcile; 297 dictionary checks across 23 operational tables; quarantines have reasons.
2. Star schema	Complete locally	0 declared orphans; employee history joins by employee_key; diagram and join map are documented.
3. Data quality	Complete locally	14 checks run on every load and each has an injection test; all 11 planted issue categories were detected.
4. Governed metrics	Complete locally	28 views return rows; catalog carries formula, source, owner and minimum group size; hand checks match.
5. Workforce dashboard	Complete locally	Month and department filters, five CEO questions, workforce cards, trend and mix charts with numeric insights.

Validation evidence

Check	Result
Automated tests	8 passed, including headcount, monthly Excel export and host-secret configuration regressions.
Rerun safety	Two full ETL refreshes produced identical row counts and business snapshots.
Metric layer	28 of 28 governed metric views return rows; M28 error is null because actuals and forecasts do not overlap.
Payroll reconciliation	18 of 18 reference months match; maximum clean payroll-to-ledger difference INR 0.00000004 (tolerance INR 0.01).
Department comparison	180 privacy-safe rows match; 18 one-person D00 rows are withheld; task differences match quarantined rows.

Raw DQ findings intentionally remain visible. The data-quality page reports REVIEW REQUIRED until an owner reviews the seeded findings; the clean-layer checks and calculations are reported separately.

Metric checks and decisions

Five independent calculations

Metric	Independent calculation	Governed view	Result
M01 headcount, D01	167	167	Pass
M07 gross pay, D01	INR 20,720,901.38	INR 20,720,901.38	Pass
M13 attendance rate, D01	96.1784%	96.1784%	Pass
M17 ledger net profit	INR 14,985,979.13	INR 14,985,979.13	Pass
M20 event-based DSO	47.9977 days	47.9977 days	Pass

Questions answered by the first dashboard

- How many people are in the workforce now, and how has headcount changed over time?

- Which departments account for the largest share of headcount?

- How many people joined and left in the selected month?

- Which location tier contains the most employees?

- Which work mode is most common?

Learning summary - owner review

The implementation shows how preserving source rows and quarantining invalid rows keeps counts explainable. Employee history must join by the version key for a time-correct department result. A governed metric view keeps dashboards and the API on one formula. A group can be analytically valid and still be unsafe to display. Confirm this summary reflects your own experience before submission.

Problems and fixes

The workforce page raised a DuckDB invalid-input error because the prior-month query had three placeholders but received two values. The department filter is now bound twice and the selected period once, with a regression test for company and department scopes. Seeded payroll, attendance, task, invoice, leave, expense, attrition-risk and SCD issues remain in raw evidence and are documented with their clean-layer handling.

Next phase and submission

The private GitHub main branch is published with the latest deployment readiness update; the phase-1 tag remains pinned to release commit 965d9f6. Still required for formal submission: deploy with production secrets, review the seeded raw-data findings, provide a changing source feed before scheduling refreshes, add fobesitech as a collaborator, verify the live link on a phone, record the five-minute demo, confirm the personal learning reflection and upload the required files.

Source: supplied fictional PulseIQ data, validation vault and project playbook. Detailed results are in docs/validation_results.md and docs/task_acceptance_matrix.md.